MILITARY SPECIFICATION

CORD ASSEMBLIES, ELECTRICAL, BRAIDED AND MOLDED JACKET

GENERAL SPECIFICATION FOR

This specification is approved for use by the Directorate of Electronic Support (AFALD/PTS), Wright-Patterson AFB, OH 45433, Department of the Air Force, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for audio frequency cord assemblies with braided and molded elastomer jackets.

1.2 Classification. The cord assemblies covered by this specification shall be of the following types:

Cord Assembly, Electrical CX-4434/U (-1) & (-2)
Cord Assembly, Electrical CX-4435/U
Cord Assembly, Electrical CX-4708A/AIC
Cord Assembly, Electrical CX-4707C/AIC

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-571	Solder, Tin Alloy, Lead-tin Alloy, and Lead Alloy

Military

MIL-I-23053/5	Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Flexible, Crosslinked
MIL-C-45662	Calibration System Requirements
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	Standard General Requirements For Electronic Equipment

DRAWINGS

Air Force

57C12661	Cord Assembly, Electrical CX-4434/U
58C12688	Cord Assembly, Electrical CX-4435/U
60C12501	Cord Assembly, Electrical CX-4707C/AIC
60C12605	Cord Assembly, Electrical CX-4708A/AIC

FSC 5995

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES INC., (UL) STANDARD

UL-94 - Standard for Flame Retardance

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville L.I., NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050 or 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Detail requirements. The cord assembly requirements shall be as specified herein and in accordance with the applicable drawings. In the event of any conflict between requirements of this specification and the drawing, the latter shall govern. (See 6.3)

3.2 First article inspection. Cord assemblies furnished under this specification shall be products that have been tested and have passed the first article inspection specified in 4.4.2, table II.

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used that will enable the cord assemblies to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Metals. The metals used shall be of proper alloy and hardness necessary to provide the required strength and rigidity with the maximum strength to weight ratio. Metals shall be of corrosion-resistant types.

3.3.2 Fungus-resistant materials. Cords shall be made of fungus-resistant materials as specified in Requirement 4 of MIL-STD-454.

3.3.3 Solder. Solder shall conform to composition Sn60 of QQ-S-571.

3.4 Design and construction.

3.4.1 Conductors. The nominal area of the conductors shall be 320 circular mils and the minimum 310 circular mils. Each conductor shall consist of 37 bunch-stranded AWG #40 (0.003145" dia) tinned cadmium-bronze wires. The strands shall be twisted to have a left hand lay of about one half inch. The individual strands, before bunching and tinning, shall have a minimum tensile strength of 90,000 pound-force per square inch and have a conductivity grade of at least 80 per cent.

3.4.1.1 Tensile strength. The tensile strength of the wire shall not decrease more than 10 per cent after tinning and insulating when tested in accordance with 4.6.1.1.

3.4.2 Insulation. The insulation shall be melt-extruded of Teflon FEP having a minimum wall thickness of .006 inches.

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3.4.3 Color code. For circuit identification, the insulation for the 6-conductor cord assembly shall be colored white, black, red, green, black, and white in that order. Insulation for 2-conductor cord assemblies shall be colored as specified on the applicable drawings. (See 2.1)

3.4.4 Stay cords. The stay cords shall be composed of extra strength rayon or polyaramid threads braided together; they shall not abrade or otherwise damage the insulation of the conductors when the cord assembly is bent or stretched.

3.4.5 Jacketing materials.

3.4.5.1 Braided. The cord shall be covered with a braided jacket composed of black nylon, 210 denier, 3 ends, 20 carrier, approx. 30 picks per inch on the six conductor cord and shall vary accordingly to give a proper fit and flexibility on the small cords and branches.

3.4.5.2 Molded elastomer. The jacket shall be a smooth, black elastomer formed free of surface irregularities. It shall be able to withstand the vertical burn test per Underwriters Laboratory Specification UL-94, class rate VO, non-dripping.

3.5 Performance.

3.5.1 Temperature range. The cord shall be flexible and resilient throughout the temperature range of -55°C to +85°C and shall not show evidence of cracking or other damage as a result of exposure to the temperature range specified herein. (See 4.6.2.1)

3.5.2 Flexing life. The 6-conductor cord assembly shall have a mean flex life of 50,000 flexes and the 2-conductor shall have a mean flex life of 25,000 flexes without showing evidence of: (See 4.6.2.2)

a. Damage to the outside jacket.
b. Damage to the insulation of the individual conductors.
c. Electrical discontinuity.

3.5.2.1 Flexing life, "Y" junction. The 6-conductor branched cord assemblies shall have a mean flexing life at the "Y" junction of 50,000 flexes and shall show no evidence of damage or discontinuity when tested according to 4.6.2.2.3, and figure 1.

3.5.3 Electrical characteristics. The electrical characteristics shall be determined on the complete cord assembly as shown on drawings 57C12661, 58C12688, 60C12501, or 60C12605.

3.5.3.1 Dielectric withstanding voltage. There shall be no evidence of breakdown when the cord assemblies are tested as specified in 4.6.2.3. Leakage currents shall be recorded and shall not vary more than 30 percent from initial values.

3.5.3.2 Insulation resistance. When tested as specified in 4.6.2.4, the insulation resistance shall not be less than 100 megohms.

3.5.3.3 Isolation. The complete cord assembly, less connector, shall meet the isolation requirements listed in Table I and when tested in accordance with figure 2 and 4.6.2.5.

TABLE I ISOLATION REQUIREMENTS

FREQUENCY Hz	ELECTROSTATIC ISOLATION (MINIMUM DECIBELS)		ELECTROMAGNETIC ISOLATION (MINIMUM DECIBELS)	
	CS-4435/U	CX-4707C/AIC	CS-4435/U	CX-4707C/AIC
20,000	115	106	86	75
10,000	121	112	92	81
5,000	127	118	98	87

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3.5.3.4 Conductor strain relief test. (Applicable to terminated cords only). A pull of 25 pounds shall be applied for a minimum of 2 minutes to the cord in the direction of the axis of connector U-174/U. A pull of 8 pounds shall be applied for a minimum of 30 minutes to the cord in the direction of the axis of connector U-173/U, U-172/U or U-179/U. When tested in accordance with 4.6.2.6.1, the cord assembly shall show no signs of jacket slippage or electrical failure and shall meet the requirements of 3.5.3.3 and 3.5.3.4.

3.5.3.5 Twist and pull. There shall be no slippage of the cord jacket within the connector twist relief clamp when tested as specified in 4.6.2.6.2. Following the test, the cord assembly shall meet the requirements of 3.5.3.3 and 3.5.3.4.

3.5.3.6 Continuity. Each conductor in the cord assembly shall be continuous when tested as specified in 4.6.2.7, and shall not exceed 4 millivolts drop above that of an equivalent length of the same conductor with a test current of 1.6 amperes (dc).

3.6 Marking. Marking shall conform to MIL-STD-130. Unless otherwise specified (See 3.1), cord assemblies shall be marked as follows:

a. Manufacturer's name or code designation.
b. Military type designation, NSN and manufacturer's part number when applicable.
c. Overall length of cord assembly where applicable.

The marking shall be on plastic bands or heat shrinkable tubing, and shall fit snugly around the cord circumference. Heat shrinkable tubing, when used, shall conform to MIL-I-23053/5, Class 1 (yellow), but shall not be used as support or part of the cord. All cord assemblies shall have one marker on the main assembly as described above or as specified on the detailed drawing. A second marker band designating a headset assembly as described in a government procurement may be placed on the main cord assembly.

3.7 Workmanship. Cord assemblies shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification of inspection. The examination and testing of cord assemblies shall be classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

(1) Inspection of product for delivery (see 4.5.1).
(2) Inspection of packaging (see 4.5.2).

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4.3 Inspection conditions. Unless otherwise specified, all inspections shall be made in accordance with the general requirements of MIL-STD-202, Section 2.

4.4 First article inspection. When required by the procurement activity, first article inspection shall be performed by the supplier, after award of contract and prior to production, at a laboratory acceptable to the Government.

4.4.1 Sample size. Ten (10) cord assemblies shall be submitted for preproduction inspection as required in table II. The ten (10) samples shall be divided into group I, consisting of five unterminated cord assemblies, and group II, consisting of five terminated cord assemblies.

4.4.2 Test routine. Sample units shall be subjected to the inspection specified in table II in the order shown, unless otherwise specified.

TABLE II FIRST ARTICLE INSPECTION

Examination or Test	Requirement paragraph	Method paragraph	Number of sample units to be inspected
Group I Unterminated			
Visual & mechanical	3.1, 3.3 3.6, 3.7	4.6.1	5
Isolation: (1) Electromagnetic (2) Electrostatic	3.5.3.3	4.6.2.5	5
Tensile strength	3.4.1.1	4.6.1.1	10 (Conductors)
Temperature range	3.5.1	4.6.2.1	5
Flexing life	3.5.2	4.6.2.2	5
Flexing life, "Y" junction	3.5.2.1	4.6.2.2.3	5
Continuity	3.5.3.6	4.6.2.7	5
Group II Terminated			
Visual & mechanical	3.1, 3.3 3.6, 3.7	4.6.1	5
Dielectric withstanding voltage	3.5.3.1	4.6.2.3	5
Insulation resistance	3.5.3.2	4.6.2.4	5
Conductor strain relief	3.5.3.4	4.6.2.6.1	5
Twist and pull	3.5.3.5	4.6.2.6.2	5
Continuity	3.5.3.6	4.6.2.7	5

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall be as specified in MIL-STD-105, and shall consist of cord assemblies covered by the drawing or procurement document employing the same type and size of connector and the same type and size of cord, produced under essentially the same conditions and offered for inspection at one time.

4.5.1.2 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. Rejected lots shall be inspected using tightened inspection.

4.5.1.3 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III.

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TABLE III GROUP A INSPECTION

Examination or test	Requirement paragraph	Method paragraph	AQL (Percent Defective) Major	Minor
Visual & mechanical	3.1, 3.3 3.6, 3.7	4.6.1		4.0%
Conductor strain relief	3.5.3.4	4.6.2.6.1	1.0%	*
Twist and pull	3.5.3.3	4.6.2.6.2	1.0%	*
Electrical				
Insulation	3.5.3.3	4.6.2.5	1.0%	**
Continuity	3.5.3.6	4.6.2.7	1.0%	**

* These mechanical defects are considered major
** All electrical defects are considered major

4.5.1.4 Sampling plan. Selection of samples for inspection shall be in accordance with MIL-STD-105 for inspection Level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.5 Group B inspection. Group B inspection shall consist of the examination and tests for flexing life (see 3.5.2 and 4.6.2.2). Two units shall be tested which have been subjected to and passed Group A inspection.

4.5.1.5.1 Sampling plan. Two cord assemblies shall be selected from the first 100 assemblies produced and from each lot of 500 or less thereafter or from every six (6) months of production, whichever occurs first. The first samples selected shall be at the start of the contract from the first inspection lot.

4.5.1.5.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.5.2 Inspection of packaging. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with section 5.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. The cord assemblies shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see table II).

4.6.1.1 Tensile strength. Wire shall meet the requirements of 3.4.1.1. Samples shall consist of one strand selected at random from each of 10 conductors of cord assemblies in group I, table II.

4.6.2 Cord assembly tests.

4.6.2.1 Temperature bend. The unterminated cord shall be placed within a test chamber wherein a temperature of 85°C (185°F) is maintained for a period of four hours. The cord shall be wrapped about a mandrel while at this temperature without damage to the cord jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cord diameter. The cord shall be attached to the mandrel and suspended vertically with the lower end weighted sufficiently to keep the specimen taut and to permit wrapping without handling. The cord shall be inspected for damage or deformation and then subjected to a test chamber wherein a temperature of minus 55°C (minus 67°F) is maintained. After 20 hours of storage at -55°C, the cord shall be wrapped about the mandrel, while at this temperature, in the same manner as it was at the positive extreme temperature. The cord shall be examined for cracks or any other damage to the jacket or conductor insulation. (See 3.5.1)

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4.6.2.2 Flexing life.

4.6.2.2.1 Six-conductor cord. Five samples of the 6-conductor unterminated cord shall be clamped and suspended through holes in a metal bar having a rectangular cross section of 1/2 inch by 5/8 inch as described below:

a. The clamping point shall be immediately above the metal bar; the cord clamp shall rest on top of the metal bar.

b. The cord shall be clamped in such a manner that it will not turn in the hole during the flexing test.

c. The length of the through holes in the metal bar shall be 5/8 inch. The diameters of the through holes shall be 0.005 to 0.010 in. greater than the diameter of the cord samples.

d. The through holes in the metal bar shall incorporate a 1/8 inch radius at the end from which the cord emerges (bending point).

e. A three-pound weight shall be hung from the cord at a point 8 to 12 inches below the bar to assure that the cord is held in a vertical position.

f. The 6 conductors of each sample cord, a small low-current drain 6-volt lamp, and a suitable power source shall be connected in series to indicate electrical continuity (see figure 3). The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetical mean of the flexing cycles of the five samples of the cord before electrical discontinuity occurs shall be taken as the flexing life of the cord. (See 3.5.2)

4.6.2.2.2 Two-conductor cord. Five samples of the 2-conductor unterminated cord shall be tested for flexing life in the same manner as the 6-conductor cord except the a one-pound weight shall be used to keep the cord taut in vertical position. (See 3.5.2).

4.6.2.2.3 Flexing life, "Y" junction. Five samples of the 6-conductor cords shall be clamped and suspended from a fixture that securely holds the "Y" junction.

a. Samples shall be clamped in the metal bar so that they are suspended entirely in the vertical position (see figure 1).

b. Samples shall be positioned so that the (Y) joint division is at the flexure point.

c. The cord shall be clamped in such a manner that it will not turn during flexing.

d. A length of nylon cord shall be tied to each of the two (Y) ends and a one-pound weight attached to the end of each (see figure 1). Polished 1/2 inch drill rods shall be attached to the flex tester so that the nylon cords will rest over them (see figure 1). The rods shall be positioned so that the (Y) joint is spread open 60 degrees (30 degrees either side of vertical).

e. The metal bar shall be supported horizontally so that it is rotated about the longitudinal axis going through the point where the sample cords emerged from the bar. The metal bar shall be rotated back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a flexing rate of approximately 85 cycles per minute.

f. The 6 conductors of each sample cord shall have a low-current drain monitor circuit (6 VDC source) connected in series to indicate electrical continuity (see figure 3). The arithmetical mean of the flexing cycles of the 5 samples before electrical discontinuity shall be taken as the flexing life. (See 3.5.2.1)

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4.6.2.3 Dielectric withstanding voltage. Dielectric strength shall be performed in accordance with Method 301 of MIL-STD-202. A potential of 600 volts, 60 Hz, shall be applied for one minute. The potential shall be applied between each conductor and the remaining conductors connected together and to the shield, and metallic shells, if present. The cord assemblies shall meet the requirements of 3.5.3.1.

4.6.2.4 Insulation resistance. Insulation resistance shall be in accordance with Method 302 of MIL-STD-202, test condition B. The potential shall be applied between each conductor and the remaining conductors connected together and to the shield, and metallic shells, if present. The cord assemblies shall meet the requirements of 3.5.3.2.

4.6.2.5 Isolation. The cord assemblies CX-4707C/AIC and CX-4435/U shall be tested for electromagnetic and electrostatic isolation to determine compliance with the limits specified herein, (see figure 2, and 3.5.3.3).

4.6.2.6 Conductor strain relief and twist relief.

4.6.2.6.1 Conductor strain relief cords assembled with connectors. The assembly shall be gripped at a point at least three inches from the connector under test in such a manner that there will be no damage to the cord or slippage in the grip. A straight pull shall be applied between the connector and the cord in a manner to avoid a sudden jerk on the connector. The cord assembly shall meet the requirements of 3.5.3.4. This test shall be performed on each end of the cord or branch that is terminated.

4.6.2.6.2 Twist and pull. With the connector held stationary, the cord shall be gripped 1/2 to 1 inch from the connector and twisted 180 degrees in a clockwise direction, released, and then twisted 180 degrees in a counter-clockwise direction and released. A 20-pound weight shall be applied gradually to the connector and held for 30 seconds. The test shall be performed on each end of the cord terminated with a connector, and the cord assembly shall then meet the requirements of 3.5.3.5.

4.6.2.7 Continuity. One and six tenths (1.6) amperes or direct current shall be passed through each conductor of each cord assembly, including end terminations. The voltage drop across the conductor and terminations shall not exceed that specified in 3.5.3.6.

5. PACKAGING. Packaging, packing, and marking shall be in accordance with MIL-C-55442.

6. NOTES.

6.1 Intended use. Cord assemblies covered by this specification are intended for use in electronic, communication, and associated electrical equipment.

6.2 Definition. An elastomer is defined as an elastic rubberlike substance such as a synthetic rubber or a plastic having some of the same physical properties of natural rubber.

6.3 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this document.

b. Title, number and date of the applicable drawing and the applicable type number.

c. Levels of preservation, packaging, packing, and the applicable marking.

6.3.1 Indirect shipment. The preservation, packaging, packing, and marking requirements specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

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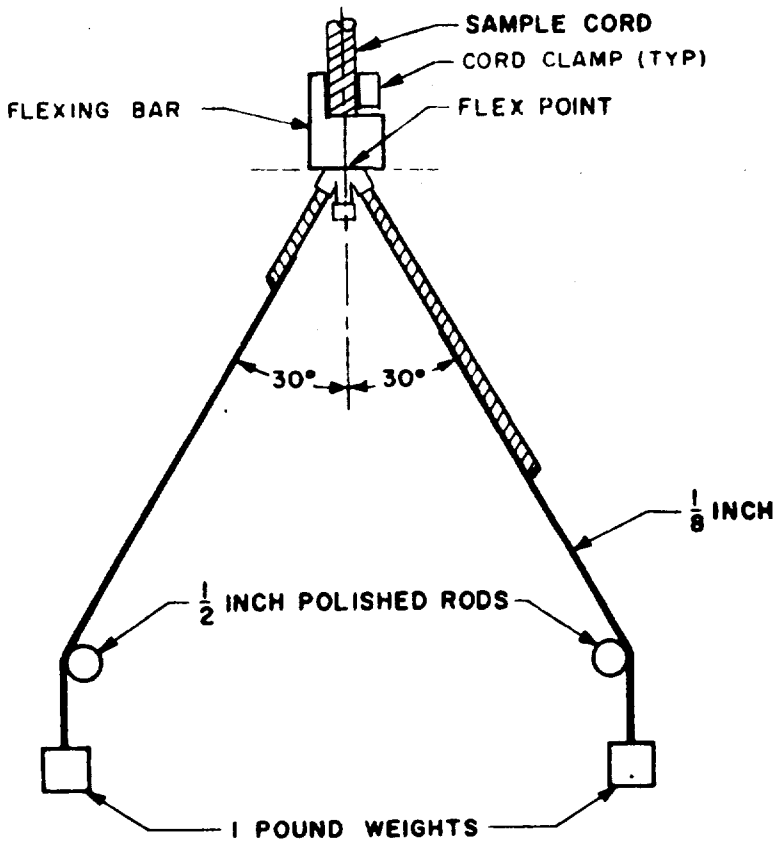
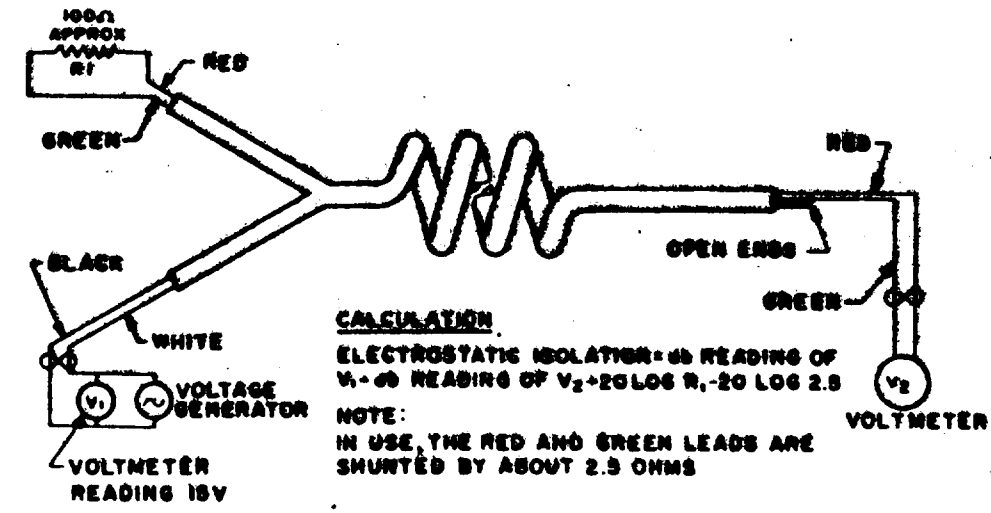
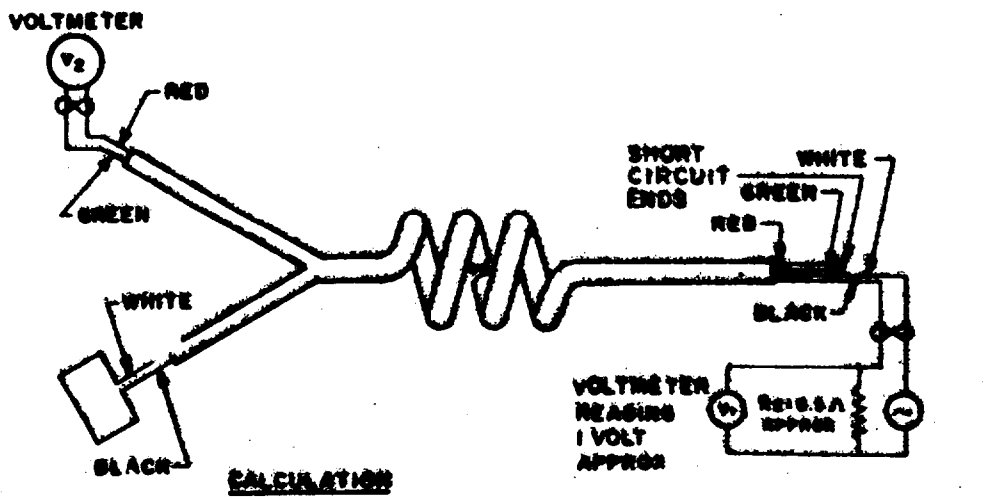


FIGURE 1 "Y" JOINT FLEXING TEST SET-UP
(See 3.7.2.1 and 4.6.2.2.3)

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ELECTROSTATIC ISOLATION TEST



ELECTROMAGNETIC ISOLATION TEST

FIGURE 2
(See 3.7.3.3 and 4.6.2.5)

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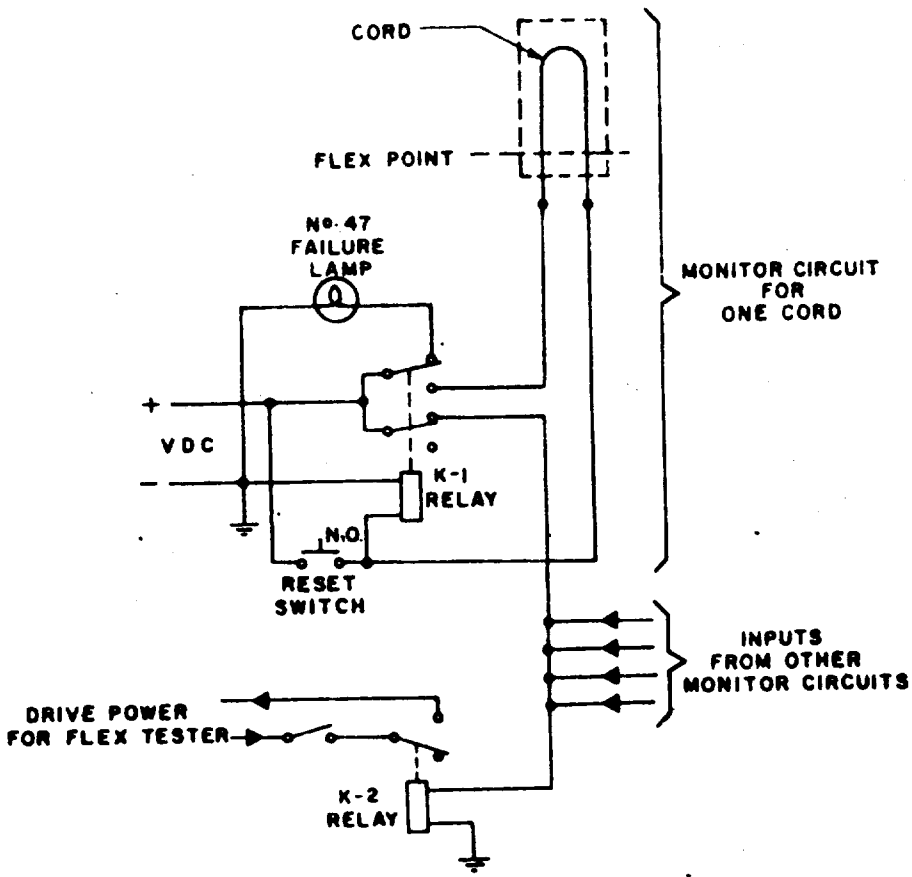


FIGURE 3 MONITOR CIRCUIT FOR CORD FLEX TESTER

(See 4.6.2.2.1 and 4.6.2.2.3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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